

Qus 201/150

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PURPOSES OF BILL OF QUANTITIES

- C.
- (i) First and foremost it enables all the Contractors tendering for a contract to price on exactly the same information with minimum effort.
 - (ii) It provides a basis for the valuation of variation which often occurs during the progress of the project work.
 - (iii) It gives an itemised list of the component parts of the building with a full description and quantity of each part, and this may assist the successful contractor in ordering materials and assessing his labour requirements for the contract.
 - (iv) After being priced it provides a good basis for a cost analysis, which subsequently will be of use on future contracts in cost planning work.



either in logical or in sequence.

finally, the computer processes the information into output sequence.

Other Services of the computer -

The computer can often speed production and reduce costs in the provision of the following services.

1. Cost planning and cost analysis
2. Bill pricing
3. the arrangement of bill information into different forms to serve other purposes.
4. financial forecasts, interim valuations and final accounts

SPECIFICATION

Definition: This is a document usually prepared by an Architect, Civil Engineer, Structural Engineer etc to supplement drawings prepared by them. It is a schedule of instructions to the Production Team Contractors.

Objectives: Its objective is to describe in a concise form, the nature and quality of materials to be used and the method of construction as well as workmanship required in carrying out the work. To amplify and explaining in detail all the issues not clear on drawings.

Types:

- (1) Descriptive Specification
- (2) Performance Specification
- (3) Proprietary Specification
- (4) Reference Specification

Descriptive Specification

It describes the materials and the standard of workmanship to be adopted in the execution of a project.

Performance Specification

It describes the functions to be performed by a proposed project. This type of specification gives the contractor the freedom to use any material and workmanship that is best suitable for the proposed project, provided the final finished work is able to perform the function expected.

Proprietary Specification

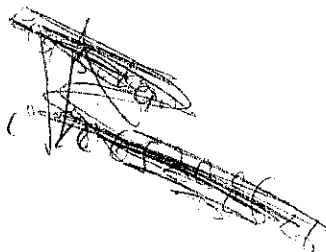
These are standards given by the manufacturers of materials, components and plant. These are usually found in Literatures or pamphlets supplied by the manufacturers

Reference Specification

It refers to the standard requirements usually prepared by the standard institutions established by Law eg. British Standard Organization provide the B.S specifications and codes of Practice, standards organizations of Nigeria (SON), The National Building Code etc.

Purpose and use of Specifications

- (a) It is a guide to bidders (i.e. Contractors) - it enables their Estimators to arrive at a fair price for the work they are bidding for
- (b) It is a contract document especially in Civil Engineering Contracts
- (c) It is a fabrication and installation guide especially for materials and components, plant and equipment, etc.
- (d) It gives in detail all the works to be done on site by a contractor



3.3 CONCISE SPECIFICATION FOR SAND AND CEMENT

SAND

Sand generally shall be clean fresh water or pit sand, free from clay, loam dirt, shells, salt or organic matter.

CEMENT

The cement shall be Portland cement complying with B.S.12. All cement shall be delivered to site in bulk cement lorries of approved design or sealed bags or in sealed tins.

3.4.1 CONCISE SPECIFICATION FOR GRAVEL AND CONCRETE WORK

GRAVEL

Coarse aggregate shall consist of fresh water river or pit gravel or of crushed granite or of equally hard and durable stone approved by the supervising officer. It shall be graded in accordance to 882 in such a way as to produce a dense and workable concrete with the minimum amount of water, to allow for adequate compaction either by hand or mechanical vibration.

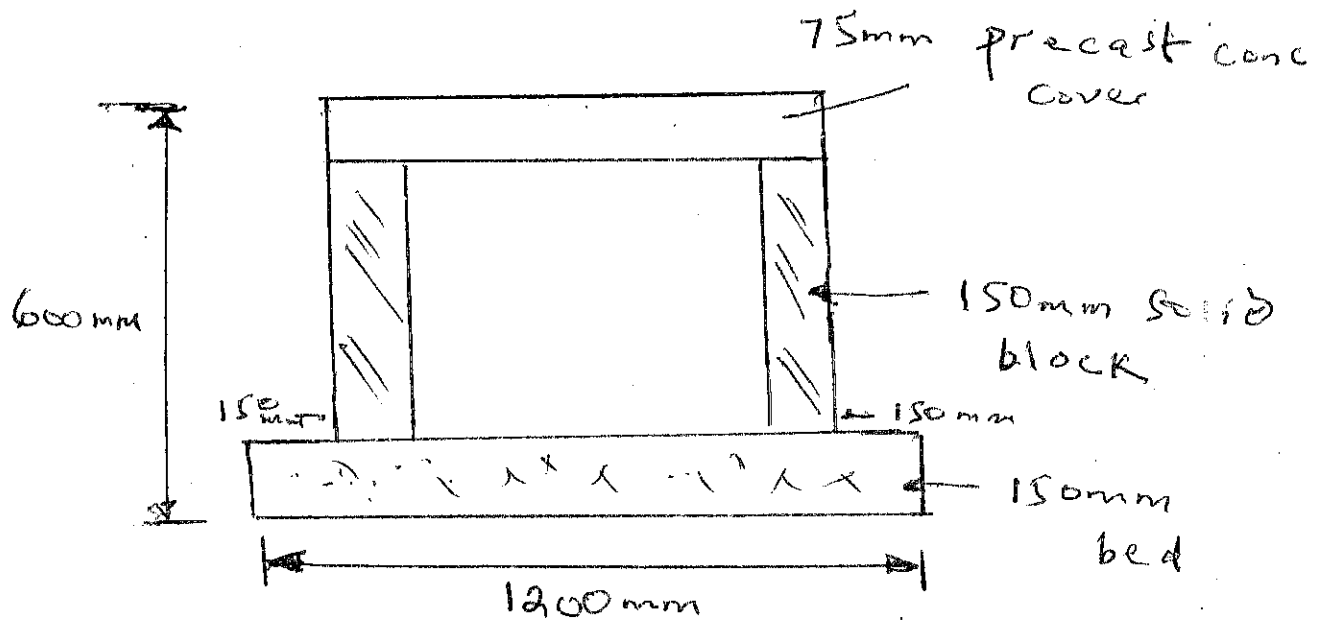
3.4.2 CONCRETE

Concrete shall be made with cement fine aggregate, coarse aggregate and water.

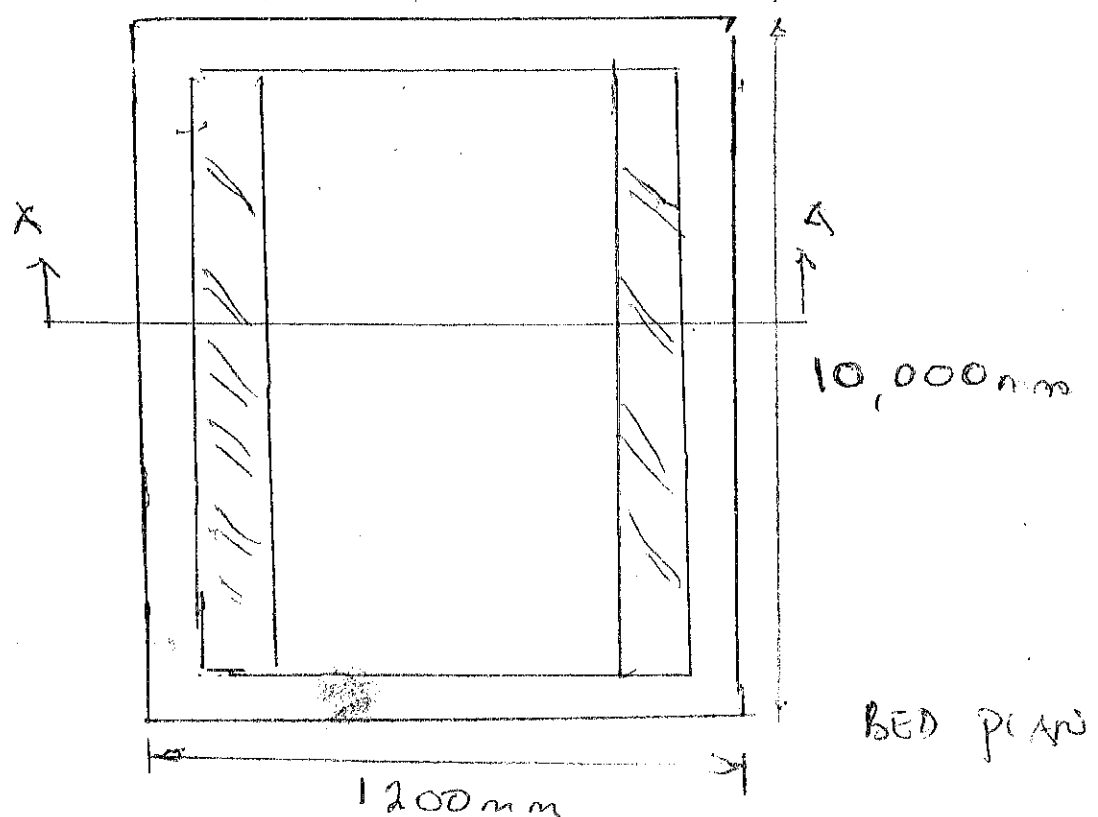
N

DRAINAGE.

Pits size - 10,000 mm length
600 mm wide
600 mm high.



Section A-X



BED PLAN

CEBOQ

A civil engineering bill of quantities is a list of items giving brief identifying descriptions and estimated quantities of the work comprised on a contract. The following are the different bill formats:

BILL FORMATS

Forms of Bills.

- (1) Work Section Bill (or Trade Bill)
- (2) Elemental Bill
- (3) Sectionalized Trade-Bills
- (4) Annotated Bill
- (5) Operational Bill
- (6) Activity Bill.

Elemental Bill

An elemental bill of quantities' is divided into appropriate building elements instead of the normal work sections .e.g. excavation and earthwork, concrete work, block work plumbing e.t.c. Within each element work may be billed in trade order or grouped in building sequence. The principal objective is to secure a more precise tendering by making the location of the work more readily identifiable and to provide a closer link with the cost plan.

Operational Bills

This type of bill subdivides the proposed project into site operations as distinct from trade or elements. That is, the site operation to be done is in executing a proposed project. Operations are defined as the work done by a man or gang of men, at some definite stage is the construction process.

Activity Bill

The activity bill is a development of the operational format but without the separation of labour and materials. It is subdivided into sections based on activities or operations derived from a network analysis. The work is measured in accordance with the prevailing standard method of measurement. Although, on site and offsite activities are usually separated and special equipment and components and the work of nominated specialist maybe grouped in separate bills. Some activities need to be completed before others can be started e.g. Foundation trenches needs to be excavated before the concrete foundations can be laid, which must in turn precedes block work.

Annotated Bills

It is possible for the bill of quantities to give the contractor the full details of the quality, type and quantity of materials and labour, and for an accurate and complete set of drawings to show him precisely where and how the work is to be executed. Nevertheless, there are always some billed items whose locations in the work is not readily identifiable and it is most useful to have a note against them in the bill giving their location. This approach has resulted in the production of annotated bills.

Annotations may be prepared in a separate document from the bill of quantities or they may be bound in at the back of the bill, although in either case they must be carefully cross-referenced.

Trade Bills/ Sectionalized Trade Bills

These are bills produced in accordance with the trade or works to be done in executing a proposed project. The trades are ascertained and their quantities determined. This is different from the elemental bill where building projects are broken down into its elements. The sectionalized trade bill was produced to overcome some of the objections to the element bill

~~Question 1~~ ~~Answer to Question 1~~

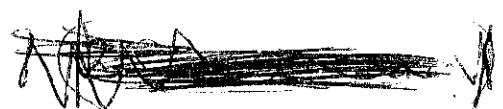
(SMM)

- A. Standard method of measurement SMM is the document of rules and provisions that could be applied in the measurement of works for all major tasks, trades, sections and elements of construction project



B. THE NEEDS FOR STANDARD METHOD OF MEASUREMENT

- (i) To facilitate pricing by standardizing the layout and content of bill of quantities
- (ii) To provide a systematic structure of bill items leading to uniform itemization and description
- (iii) To provide a rational system of billing suitable for both manual and computers operation
- (iv) To simplify the measurement of works and administration of contracts
- (v) To provide a uniform basis for measuring the works so as to avoid misunderstanding and ambiguities
- (vi) To assist in the financial control of the work



B.

i) Taking off:

The dimensions are measured scaled or read from drawings and entered in a recognise form or specially ruled paper, called dimension paper

ii) Abstracting

The various items are sorted out into their respective trades as they are entered on the abstract paper. They are arranged in the following sequences (a) cube (b) superstructure (c) linear (d) numbers

iii)

Direct billing

For a normal simple domestic or small industrial building items taken off may not require to be abstracted. The bill of quantities could be written direct from the dimension after squaring, but this needs a lot of professional skill and experience



~~ANSWER TO QUESTION 201/08~~

C.

Timing column
Dimension column
Squaring column

Description column

